CareAI — Diabetes Risk Prediction System

System Architecture Specification & Technical Documentation — Category: Predictive Healthcare & Machine Learning

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1. Clinical Background & Screening Rationale

Type 2 diabetes mellitus often progresses asymptomatically for years before clinical diagnosis, leading to irreversible cardiovascular, renal, and retinal complications. CareAI provides an accessible, non-invasive screening platform using canonical metabolic markers to compute probabilistic risk tiers (Low, Moderate, High) and provide early lifestyle intervention guidelines.

2. Dataset & Clinical Input Parameters

Feature	Clinical Measurement	Standard Healthy Reference Range
Pregnancies	Count of pregnancies	0 for males / null applicable
Glucose	Plasma glucose (mg/dL)	70 - 99 mg/dL (fasting normal)
Blood Pressure	Diastolic BP (mm Hg)	< 80 mm Hg (normal)
Skin Thickness	Triceps fold thickness (mm)	10 - 25 mm
Insulin	2-Hour serum insulin (μU/mL)	16 - 166 μU/mL
BMI	kg / m²	18.5 - 24.9 (normal weight)
DPF	Diabetes Pedigree Function	Genetic family hereditary risk score
Age	Chronological age (years)	Adult patient age

3. Machine Learning Architecture & Evaluation Metrics

• **Preprocessing:** Numerical feature standardization via Scikit-learn `StandardScaler` to equalize metric variances.

• **Classifier:** `RandomForestClassifier` ensemble with 100 estimators, Gini impurity splitting, and bootstrap sampling.

• **Performance:** ~88% classification accuracy, 80% precision, 76.5% recall, 78.2% F1-score, and 95.3% ROC-AUC on hold-out validation splits.

• **Feature Importance:** Gini importance ranking identifies Plasma Glucose, BMI, and Age as the three primary diagnostic drivers.

4. Application Interface & Analytics

• **Streamlit UI:** Patient demographic inputs, sample profile auto-fill buttons (Healthy, Prediabetic, High Risk), and color-coded risk cards.

• **Persistent Analytics:** Historical screening log with CSV export, interactive Matplotlib/Seaborn donut charts, and ADA/WHO guidance notes.